

# Vide Game AI - Decision Tree CRUD for NPC Behavior

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# Introduction

- **NPCs (Non-Player Characters) make decisions in games**
- **Actions include:**
  - **Attack**
  - **Patrol**
  - **Run**
  - **Search**
- **Decision Trees help simulate intelligent behavior**

# **Problem Statement**

**NPCs need smart decision-making**

**Manual logic is complex**

**Need system to:**

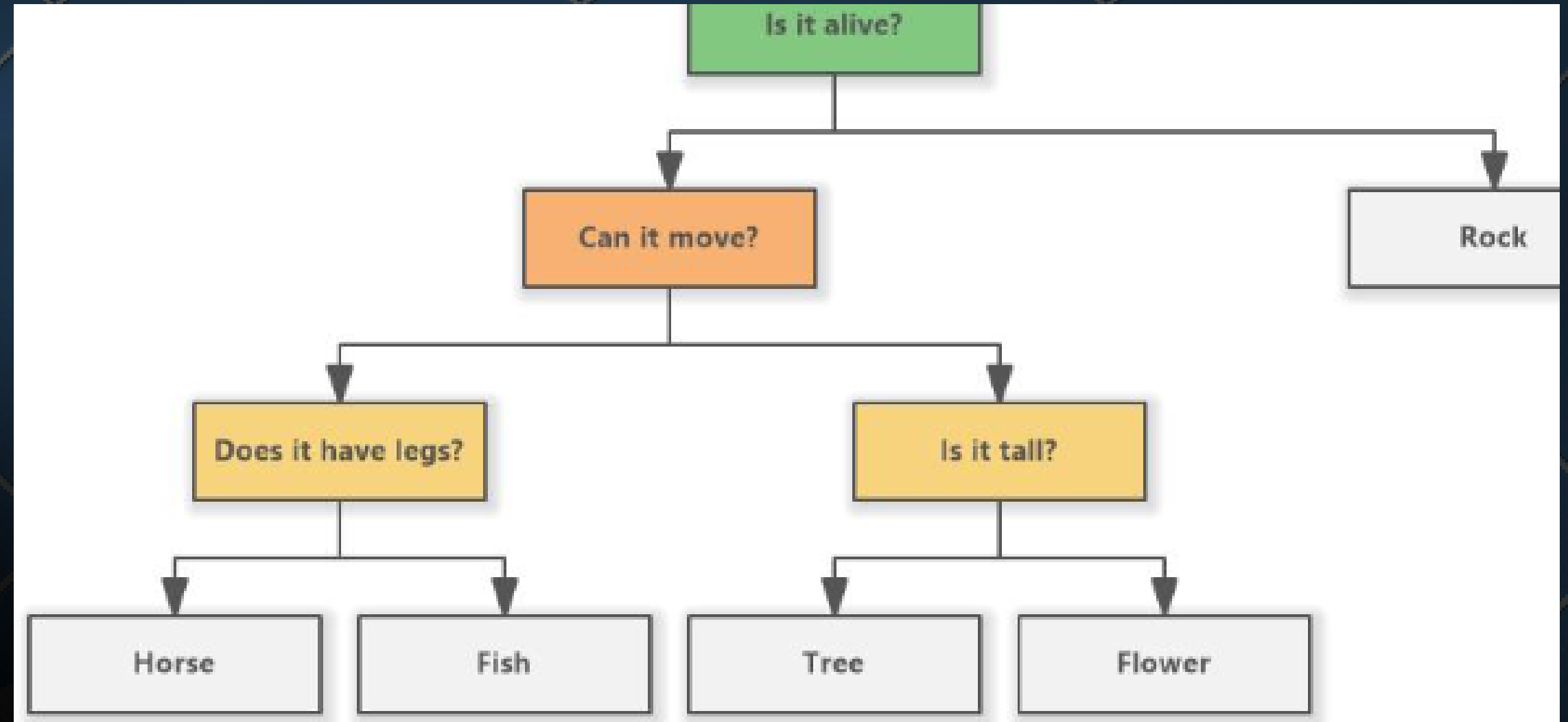
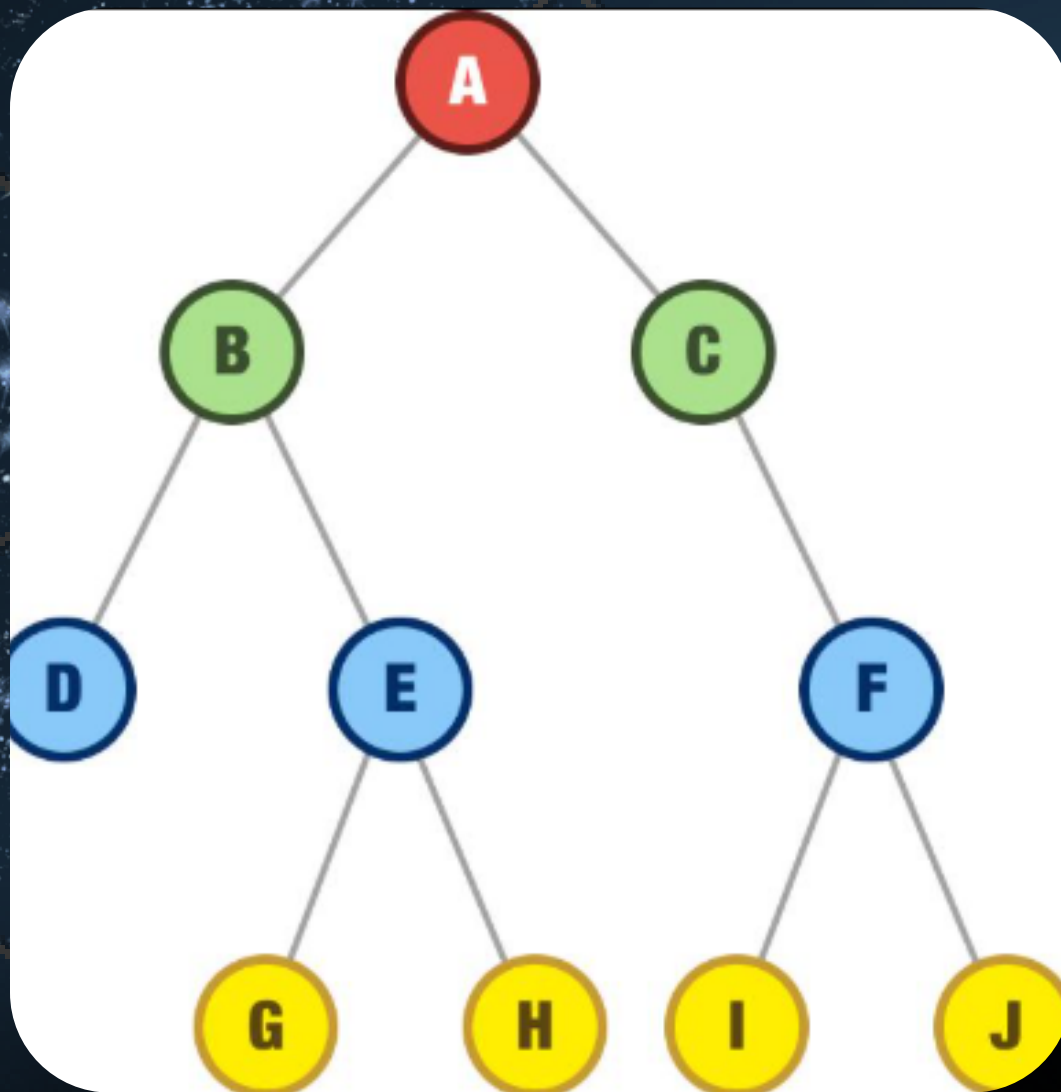
**Add decisions**

**Update decisions**

**Delete decisions**

**Search decisions**

**Solution: Decision Tree + CRUD Operation**



Data Structure Used

Binary Tree

Node contains:

ID

Decision

Left child

Right child

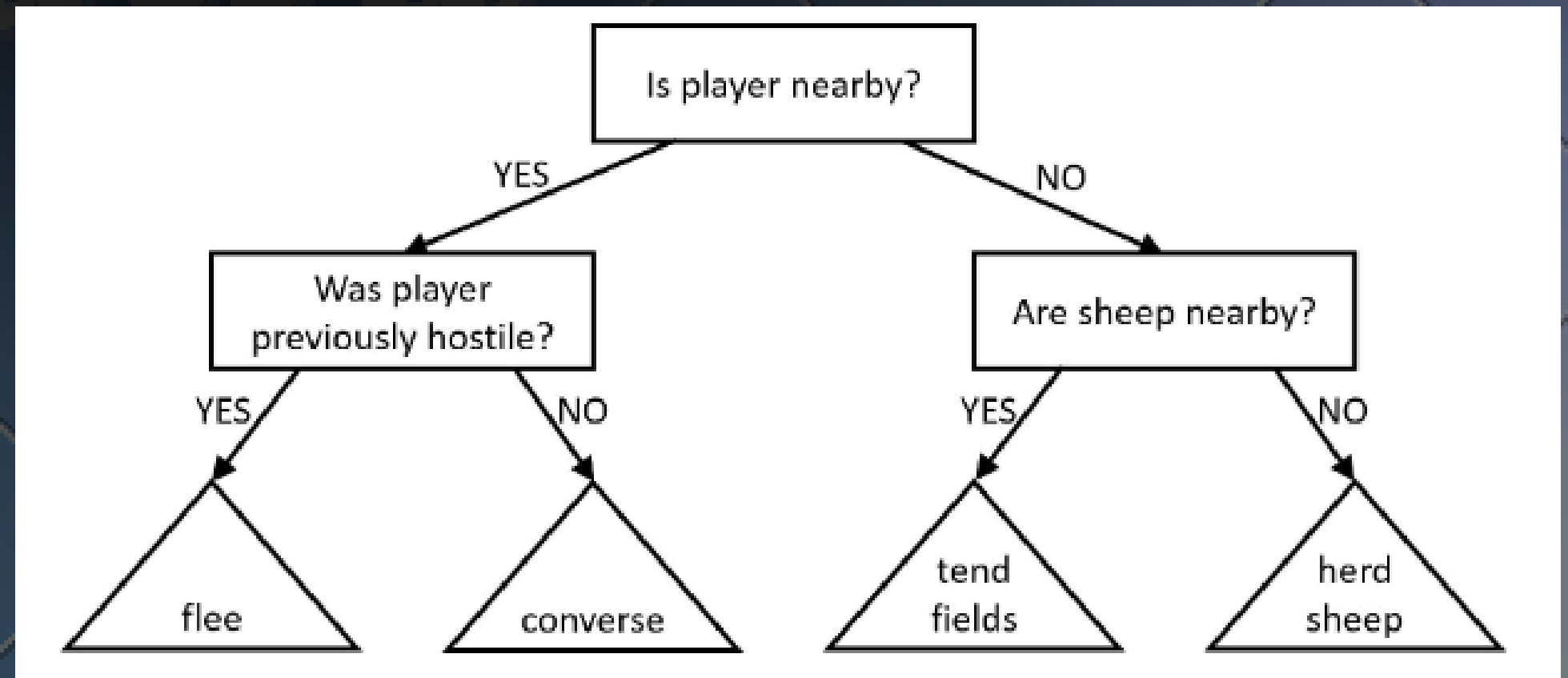
# Node Structure (Code)

```
struct Node
{
    int id;
    char decision[100];
    struct Node *left;
    struct Node *right;
};
```

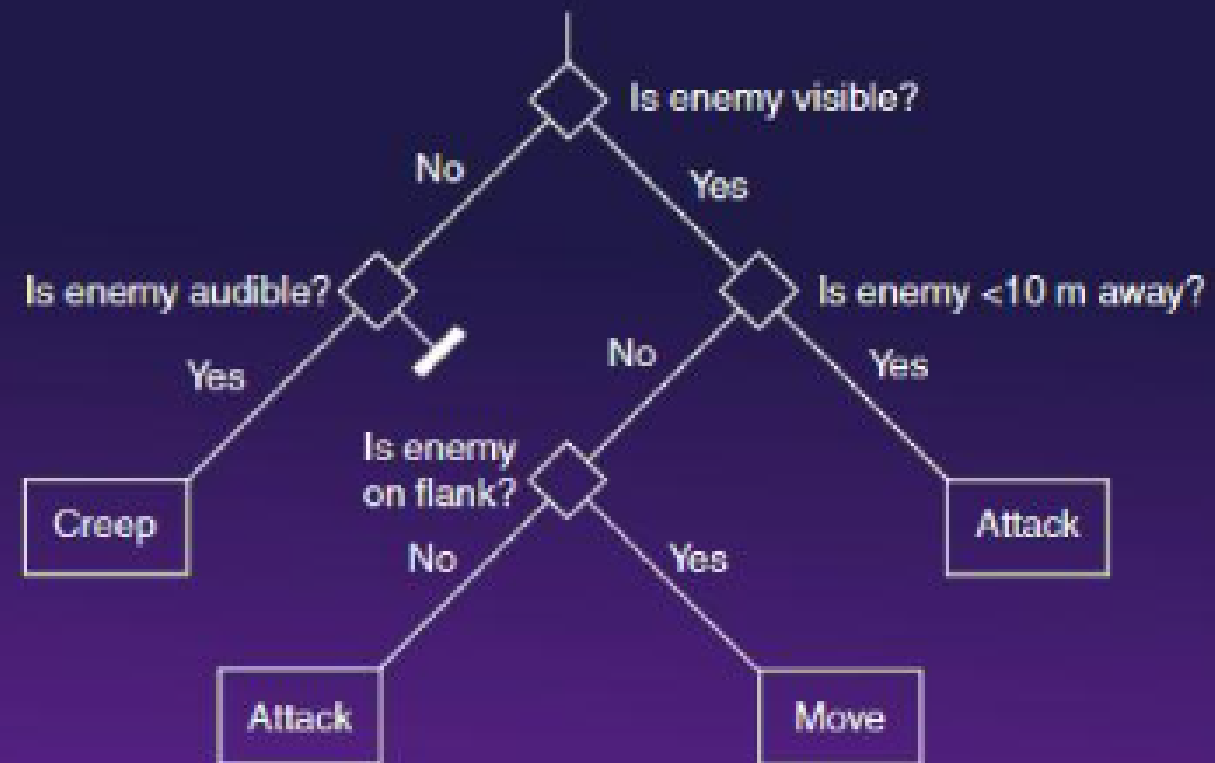
## System Design

Example Decision Tree:  
Enemy Nearby?

Yes → Attack Enemy  
No → Patrol Area

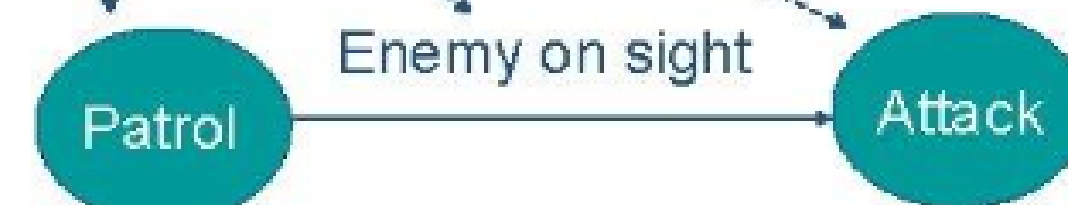


## Decision Tree

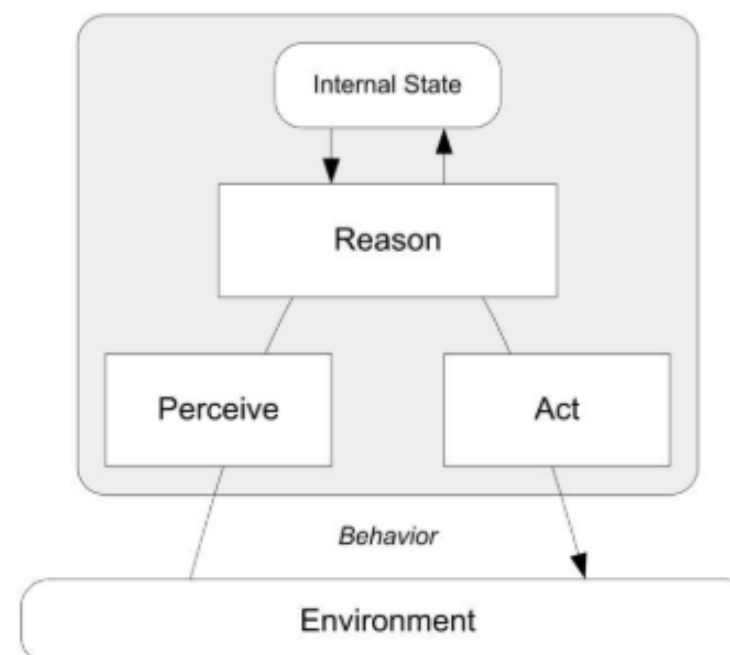


## NPC's Behavior is Modeled through Finite State Machines (I)

- **State**: an activity performed by an avatar
- **Event**: something that happens in the game world that makes state change



## NPC Behavior



## CRUD Operations

- Create → Add Node
- Read → Display Tree
- Update → Modify Decision
- Delete → Remove Node

### Algorithm (Insert)

1. Start
2. Check if tree empty
3. Create node
4. Compare IDs
5. Insert left or right

## Algorithm (Search)

1. Compare ID with root
2. If equal → Found
3. If smaller → Left
4. If larger → Right

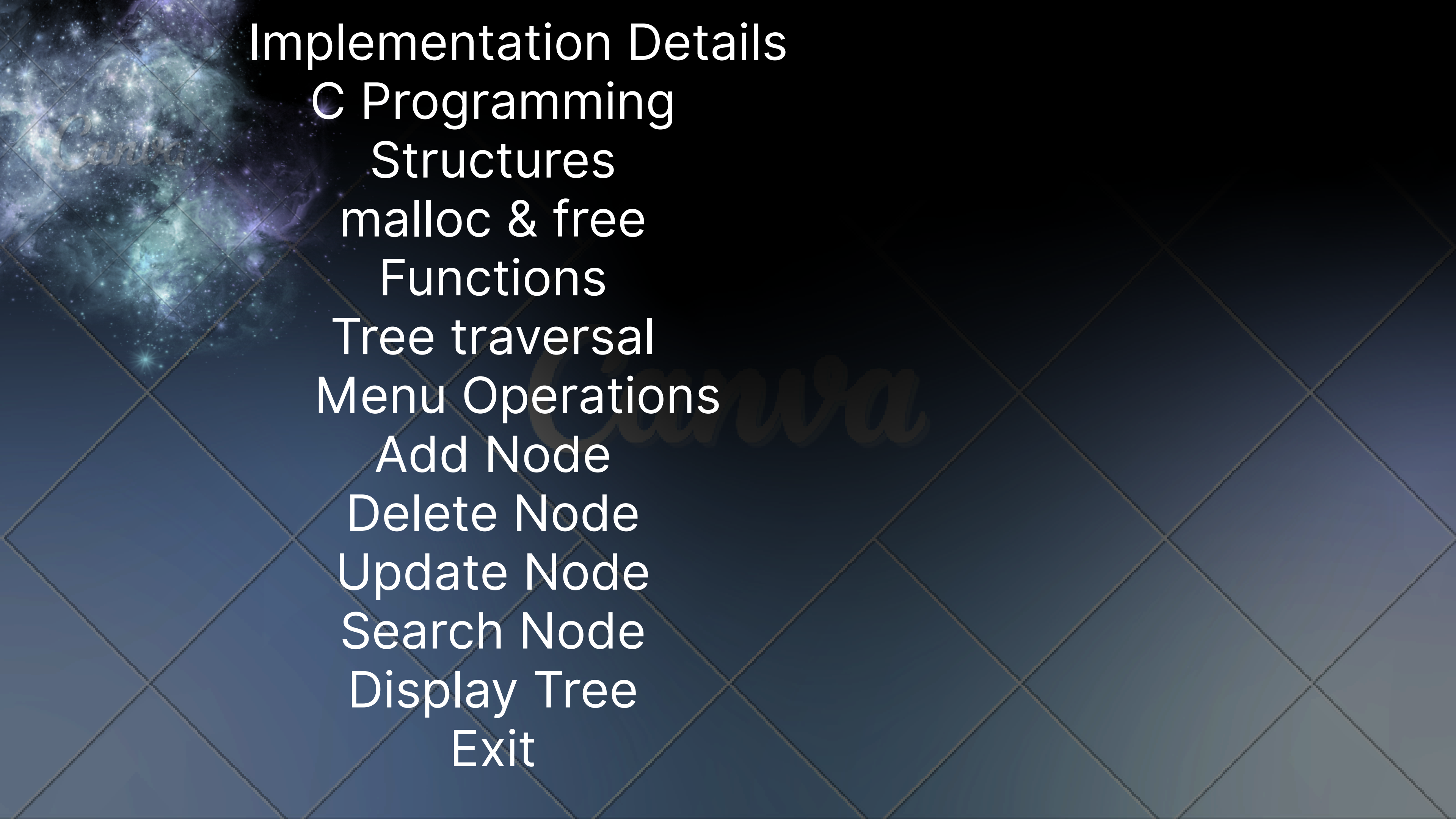
## Algorithm (Update & Delete)

### Update:

- Search node
- Modify decision

### Delete:

- Find node
- Remove it
- Adjust tree



# Implementation Details

C Programming

Structures

malloc & free

Functions

Tree traversal

Menu Operations

Add Node

Delete Node

Update Node

Search Node

Display Tree

Exit

# Sample Output

===== VIDEO GAME AI =====

1. Add Node

2. Display Tree

Node ID: 10

Decision: Attack Enemy

Advantages

Simple decision making

Easy to modify

Efficient structure

Real-world application

# Limitations

- Console-based only
- No graphical view
- Basic AI logic

# Future Enhancements

- Graphical tree visualization
- Game engine integration
- Advanced AI logic
- Multiple conditions



# Conclusion

- Implemented Decision Tree
- CRUD operations successful
- Useful for Game AI
- Learned data structures & memory

# Thank you